

To chalo, shuru karte hain error and measurement... Sach bataun to is chapter mein physics bahut kam hai, aur maths aur measurement zyada hai.

Lekin exam ke POV se yeh bahut achha chapter hai, kyunki isse regular basis par questions aate hain aur score karne wale hote hain...

Lets start



SIGNIFICANT FIGURE

- ★ Significant figures are the digits in a number that contribute to its precision and are considered reliable. They indicate the precision of a measurement or calculation and are crucial for conveying the reliability of a numerical value.
- ★ Larger the no. of significant figure obtained in a measurement larger will be its accuracy.

Rules

- ★ All the non-zero digits are significant.
Ex $2573 \rightarrow (4)$
- ★ All the zeros between two non-zero digits are significant.
Ex $200703 \rightarrow (6)$
- ★ All the leading zeros are insignificant (zero before 1st non-zero digit)
Ex $.0023 \rightarrow (2)$
 $00235 \rightarrow (3)$
- ★ The trailing zeros in a number with decimal points are significant. (zeros to the right of last non-zero digit one significant)
 $1.020 \rightarrow 4$
 $1.02000 \rightarrow 6$

- ★ The trailing zeros in a number without a decimal point are not significant.

$$35000 \rightarrow 2$$

$$14200 \rightarrow 3$$

- ★ Changing the units does not change the no. of significant digits.

$$5.03 \text{ m} \rightarrow (3)$$

$$503 \text{ cm} \rightarrow (3)$$

- ★ Number of Significant digit does not depends on system of Units and it also does not change with scientific notation

$$32 \times 10^5 \rightarrow (2)$$

$$320 \times 10^4 \rightarrow (2)$$

$$3.2 \times 10^6 \rightarrow (2)$$

Rules for Rounding Off Digits:

Sabse pahle ye dekho ki kitne significant digit me round off karna hai matlab kise rakhna kise bhagana hai.



To represent the result of any computation containing more than one uncertain digit, it is rounded off to appropriate number of significant figures.

Rules for Rounding Off the Numbers:

- Rule:** If the digit to be rounded off is more than 5, then the preceding digit is increased by one. e.g. $6.87 \approx 6.9$
- Rule:** If the digit to be rounded off is less than 5, then the preceding digit is unaffected and is left unchanged. e.g. $3.94 \approx 3.9$
- Rule:** If the digit to be rounded off is 5 then the preceding digit is increased by one if it is odd and is left unchanged if it is even. e.g. $14.35 \approx 14.4$ and $14.45 \approx 14.4$

Q. Round off following number into 3 significant figure.

Sol. In the following example 7 is an odd number so it's value will increase by 1 according to the rules stated for rounding off.

$$\begin{array}{l}
 \begin{array}{c} +1 \\ \curvearrowright \\ 1.8762 \rightarrow 1.88 \\ 1.8769 \rightarrow 1.88 \\ 1.8758 \rightarrow 1.88 \\ 1.8752 \rightarrow 1.88 \\ 1.8751 \rightarrow 1.88 \end{array} \\
 \begin{array}{c} 1.8750 \rightarrow 1.88 \\ \downarrow \text{odd} \quad \downarrow \text{even} \\ 1.8850 \rightarrow 1.88 \\ \downarrow \text{even} \quad \downarrow \text{even} \end{array} \\
 \begin{array}{l} 1.8732 \rightarrow 1.87 \quad 1.876 \rightarrow 1.88 \\ 1.8749 \rightarrow 1.87 \quad 1.87621 \rightarrow 1.88 \\ 1.875001 \rightarrow 1.88 \end{array}
 \end{array}$$

Q. Round off following to 3 Significant digit.

$$\begin{array}{l}
 2.3572 \rightarrow 2.36 \\
 2.3758 \rightarrow 2.38 \\
 5.7342 \rightarrow 5.73 \\
 8.00245 \rightarrow 8.00 \\
 1367 = 1.367 \times 10^3 \rightarrow 1.37 \times 10^3 = 1370
 \end{array}$$

Q. Keep only 3 Significant figure in the following number:

- (1) $1.5587 \rightarrow 1.56$
- (2) $0.0037421 \rightarrow 0.00374$
- (3) $1.6683 \times 10^6 \rightarrow 1.67 \times 10^6$
- (4) $13.67 \rightarrow 13.7$
- (5) $128522 \rightarrow 129 \times 10^3$
- (6) $5867523 \rightarrow 587 \times 10^4$

Rules for Addition/Subtraction

★ In addition or subtraction final result should retain as many decimal places as are there in the number with the least decimal places.

★ **Rule-1:** In multiplication or division, the final result should retain as many significant figures as are there in the original number with the least significant figures.

Jin numbers ko jod rahe hain, unmein yeh dekho ki decimal ke baad kiske kitne digits hain.

Humein final answer round off karke aisa answer dena hai, jisme decimal ke baad utne hi digits ho, jitne add/subtract hone wali numbers mein minimum hain.



Ex: Add following in correct no. of significant figure.

$$\begin{array}{r}
 3.478 \quad 3 \\
 + 4.2 \quad 1 \\
 \hline
 7.678 \text{ (calculator ans)}
 \end{array}$$

Ans = after decimal 1 digit form

But Ans $\rightarrow 7.7$ (using round off)

Ex: Multiply following in correct number of significant figures.

$$6.321 \times 5.87 = 37.10427 \text{ (calculator ans.)}$$

4 sig. fig 3 sig. fig

So, our answer will be in 3 sig. fig hence, round off the calculator ans. to 37.1

$$\begin{array}{r}
 3.734 \\
 \times 4.32 \\
 \hline
 16.13088
 \end{array}$$

Sol. = 16.1

$$\begin{array}{r}
 2.4732 \\
 1.834 \\
 + 2.03 \\
 \hline
 6.3372
 \end{array}$$

Sol. = 6.34

$$\begin{array}{r}
 5.9987 \\
 - 2.07 \\
 \hline
 3.9287
 \end{array}$$

Sol. = 3.93

Q. If $R_1 = 5.272$

$$R_2 = 3.41$$

Find Req if they are connected in series

$$\begin{array}{r}
 5.272 \quad 3 \\
 3.41 \quad 2 \\
 \hline
 8.682
 \end{array}$$

Neeche bahut saare questions attach kar raha hoon. Exam se just pehle inhe upar-upar se dekh lena, mast revision ho jaayega.



Practice

Calculation	Calculator says	Answer
3.24 m + 7.0 m	10.24 m	10.2m
100.0g - 23.73g	76.27 g	76.3g
0.02cm + 2.371m	2.391 cm	2.39 cm
713.3 L - 3.872 L	709.228 L	709.2 L
1818.2 lb. + 3.37 lb	1821.57 lb	1821.6 lb.ft
1.030 g × 2.87 mL ④ ③	2.9561 g/mL	2.96g/mL

MULTIPLICATION AND DIVISION

★ **Rule - 2:** In carrying out a multiplication or division, the answer cannot have more significant figures than either of the original numbers.

$$\frac{23.7606838 \text{ kg}}{11.70 \text{ m}^3} = 2.03082767 \text{ kg/m}^3$$

Three significant figures Calc Three significant figures

Four significant figures

Practice

Calculation	Calculator says	Answer
3.24m × 7.0 m	22.68 m ²	23m ²
100.0g ÷ 23.7 cm ²	4.219409283 g/cm ³	4.22 g/cm ³
0.02cm × 2.371 cm	0.04742 cm ²	0.05 cm ²
710m ÷ 3.0 s	26.6666667 m/s	240 m/s
1818.2lb × 3.23 ft	5872.786 lb.ft	5870 lb.ft
1.030g × 2.87 mL	2.9561 g/mL	2.96 g/mL

Q. A ohm's law experiment, reading of voltmeter across the resistor is 12.5 V and reading of current $I = 0.20$ Amp. Estimate the resistance in correct S.F.?

Sol. $V = iR$

$$v = 12.5$$

$$i = 0.20$$

$$R = \frac{v}{i} = \frac{12.5}{0.20} = 62.5$$

3 sig. fig Round off 62

2 sig. fig

Ans: 62 Ω

ERRORS IN MEASUREMENT

Error in measurement refers to the difference between the true value and the measured value of a quantity is known as the error of measurement.

Classification of Errors



Errors may arise from different sources and are usually classified as follows:

- 1. Systematic or controllable errors:** Systematic errors are the errors whose causes are known. They can be either positive or negative. Systematic errors can further be classified into three categories:
 - (i) Instrumental errors:** These errors are due to imperfect design or erroneous manufacture or misuse of the measuring instrument. These can be reduced by using more accurate instruments.
 - (ii) Environmental errors:** These errors are due to the changes in external environmental conditions such as temperature, pressure, humidity, dust, vibrations or magnetic and electrostatic fields.
 - (iii) Observational errors:** These errors arise due to improper setting of the apparatus or carelessness in taking observations. This can be reduced by proper setting of the instrument before we start using it.

(iv) **Personal Error:** This type of error may creep if the experimenter is not careful enough while performing the experiment and noting down the readings.

2. **Random errors:** These errors are due to unknown causes. Therefore they occur irregularly and are variable in magnitude and sign. Since the causes of these errors are not known precisely they can not be eliminated completely. For example, when the same person repeats the same observation in the same conditions, he may get different readings different times.

Random errors can be reduced by repeating the observation a large number of times and taking the arithmetic mean of all the observations. This mean value would be very close to the most accurate reading.

ERROR OF MEASUREMENT

Let $a_1, a_2, a_3, a_4, \dots, a_n$ be the experimental values or measured values of a quantity a . Mean value is taken as the true value when true value is not given.

$$\text{True value} = a_m (\text{let}) = \text{Mean Value} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

$$\text{Absolute error in Exp no. 1} = |a_m - a_1| = |\Delta a_1|$$

$$\text{Absolute error in Exp no. 2} = |a_m - a_2| = |\Delta a_2|$$

$$\text{Absolute error in Exp no. 3} = |a_m - a_3| = |\Delta a_3|$$

$$\text{Absolute error in Exp no. } n^{\text{th}} = |a_m - a_n| = |\Delta a_n|$$

Mean absolute error =

$$\frac{|\Delta a_1| + |\Delta a_2| + |\Delta a_3| + \dots + |\Delta a_n|}{n} = \Delta \bar{a}_m$$

$$\text{Representaion} \Rightarrow (a_m \pm \Delta \bar{a}_m)$$

$$\text{fractional error or Relative error} = \frac{\Delta \bar{a}_m}{a_m}$$

$$\% \text{ error} = \frac{\Delta \bar{a}_m}{a_m} \times 100$$

Q. In calculation of value of experimentally result is represented as $g = 10 \pm .2$. Find fractional error or relative error

Sol. $g = 10 \pm .2$

↓
True Value

$$\text{fractional error relative error} = \frac{.2}{10}$$

$$\% \text{ error} = \frac{.2}{10} \times 100 = 2\%$$

Q. In the exp. for calculation of g value of g is found as 10.2, 10, 9.8, 10.5, 9.5. Find

1. True value
2. Mean absolute error
3. Relative error
4. Percentage error

Sol. (1) Find true value (mean value) = g_m

$$= \frac{10.2 + 10 + 9.8 + 10.5 + 9.5}{5} = \frac{50}{5} = 10.0$$

(2) mean absolute error

$$\frac{|10 - 10.2| + |10 - 10| + |10 - 9.8| + |10 - 10.5| + |10 - 9.5|}{5}$$

$$\Delta \bar{g}_m = \frac{.2 + 0 + .2 + .5 + .5}{5} = \frac{1.4}{5} = .28$$

$$g \Rightarrow 10.0 \pm .28$$

$$(3) \text{ Relative error} = \frac{.28}{10}$$

$$(4) \% \text{ error} = \frac{.28}{10} \times 100$$

Q. The time period of oscillation of a simple pendulum in an experiment is recorded as 2.63 s, 2.56 s, 2.42 s, 2.71 s and 2.80 s respectively. Find (i) mean time period (ii) absolute error in each observation and percentage error.

Sol. (i) Mean time period is given by

$$\begin{aligned} \bar{T} &= \frac{2.63 + 2.56 + 2.42 + 2.71 + 2.80}{5} \\ &= \frac{13.12}{5} = 2.62\text{s} \end{aligned}$$

(ii) The absolute error in each observation is

$$2.62 - 2.63 = -0.01,$$

$$2.62 - 2.56 = 0.06,$$

$$2.62 - 2.42 = 0.20,$$

$$2.62 - 2.71 = -0.09,$$

$$2.62 - 2.80 = -0.18$$

$$\text{Mean absolute error, } \overline{\Delta T} = \frac{\sum |\Delta T|}{5}$$

$$= \frac{0.01 + 0.06 + 0.2 + 0.09 + 0.18}{5} = 0.11\text{sec}$$

$\therefore$ Percentage error is

$$= \frac{\overline{\Delta T}}{\bar{T}} \times 100 = \frac{0.11}{2.62} \times 100 = 4.2\%$$

PROPAGATION OF ERROR IF PHYSIC QUANTITY ARE ADDED OR SUBTRACTED

$$Y = A + B, (\text{let})$$

$$A = a \pm \Delta a$$

$$B = b \pm \Delta b$$

$$A + B = (a + b) \pm (\Delta a + \Delta b)$$

♦ $A = 20.3 \pm .2$

$B = 14.1 \pm .3$

find $A + B = ?$

$$Y = (A + B) = 34.4 \pm .5$$

♦ $R_1 = 5.6 \pm .2$

$R_2 = 7.4 \pm .3$

Find Req if R_1, R_2 are in series

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = 13.0 \pm .5$$

$$\% \text{ error in } R_{eq} = \frac{.5}{13} \times 100$$

Substraction

$$A = a \pm \Delta a$$

$$B = b \pm \Delta b$$

$$A - B = (a - b) \pm (\Delta a + \Delta b)$$

→ '-' nahi aata.

$$A = a \pm \Delta a$$

$$B = b \pm \Delta b$$

$$A + B = (a + b) \pm (\Delta a + \Delta b)$$

$$A - B = (a - b) \pm (\Delta a + \Delta b)$$

Ex.

$$A = 57.8 \pm .6$$

$$B = 25.3 \pm .2$$

$$A - B = 32.5 \pm (.8)$$

$$A + B = 83.1 \pm .8$$

Q. If initial temp is $(57.3 \pm .8)^\circ\text{C}$

And final temp is $(89.7 \pm .5)^\circ\text{C}$

Calculate the rise in temperature with proper error limits.

Sol. $T_f - T_i = (32.4 \pm 1.3)^\circ\text{C}$

★ If $y = \frac{ab^2c^3}{d^4}$

$$\left(\frac{\Delta y}{y}\right)_{\text{max permixable error}} = \frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 3\frac{\Delta c}{c} + 4\frac{\Delta d}{d}$$

★ $z = \frac{a^3b^5}{c^2}$

$$\frac{\Delta z}{z} = 3\frac{\Delta a}{a} + 5\frac{\Delta b}{b} + 2\frac{\Delta c}{c}$$

Q. If $y = \frac{a^2b^3}{c^4}$ if fractional error of a, b, c are .2.

Sol. .1, .3 respectively find fractional error in

$$y = \frac{\Delta y}{y} = ?$$

$$\frac{\Delta y}{y} = 2\frac{\Delta a}{a} + 3\frac{\Delta b}{b} + 4\frac{\Delta c}{c}$$

$$? = 2 \times .2 + 3 \times .1 + 4 \times .3 = .4 + .3 + 1.2 = 1.9$$

Yeh ek bahut important sawaal hai jo har saal Mains mein aata hai, isliye iski practice achhe se kar lena. Main yahin thode zyada questions attach kar raha hoon.



Q. If $y = \frac{a^3b^2}{c^5}$ if % error in a, b, c are .5%, .3%, .2% respectively

Find % error in y .

Sol. $\frac{\Delta y}{y} = 3\frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 5\frac{\Delta c}{c}$

$$\frac{\Delta y}{y} \times 100 = \frac{3\Delta a}{a} \times 100 + \frac{2\Delta b}{b} \times 100 + \frac{5\Delta c}{c} \times 100$$

$$\frac{\Delta y}{y} \times 100 = 3 \times .5 + 2 \times .3 + 5 \times .2$$

$$\% \text{ error in } y = 1.5 + .6 + 1.0 = 3.1\%$$

Q. A physical quantity X is given by $X = \frac{2k^3l^2}{m\sqrt{n}}$. The

percentage error in the measurements of k, l, m and n are 1%, 2%, 3% and 4% respectively. The value of X is uncertain by

Sol. $x = 2k^3l^2m^{-1}n^{-\frac{1}{2}}$

$$100 \times \frac{\Delta x}{x} = 3 \times 1 + 2 \times 2 + 1 \times 3 + \frac{1}{2} \times 4 = 12$$

Q. A physical quantity z depends on four observables

a, b, c and d , as $Z = \frac{a^2 b^3}{\sqrt{cd^3}}$. The percentages of

error in the measurement of a, b, c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is:

Jee Main 2020 (Online) 5th Sept., Morning Shift

Sol. $\frac{\Delta z}{z} \times 100 = 2 \times 2 + \frac{2}{3} \times \frac{3}{2} + \frac{1}{2} \times 4 + 3 \times 2.5$
 $= 4 + 1 + 2 + 7.5 = 14.5$

Q. The percentage errors in quantities P, Q, R and S are 0.5%, 1%, 3% and 1.5% respectively in the measurement of a physical quantity

$A = \frac{P^3 Q^2}{\sqrt{RS}} = P^3 Q^2 R^{-\frac{1}{2}} S^{-\frac{1}{2}}$. The maximum percentage

error in the value of A will be :

Jee Main 2018 (Online) 16th April, Morning Shift

Sol. $3 \times \frac{\Delta P}{P} + 2 \times \frac{\Delta Q}{Q} + \frac{1}{2} \times \frac{\Delta R}{R} + \frac{1}{2} \times \frac{\Delta S}{S}$
 $3 \times .5 + 2 \times 1 + \frac{1}{2} \times 3 + \frac{1}{2} \times 1.5$
 $= 1.5 + 2 + 1.5 + 1.5$



Kaam ka Dabba

★ For a simple pendulum,

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\left(\frac{\Delta T}{T} \right)_{\text{max. possible error}} = \frac{1}{2} \frac{\Delta l}{l} + \frac{1}{2} \frac{\Delta g}{g}$$

$$T \propto l^{1/2} \text{ (if } g \text{ is constant)}$$

$$\Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l}$$

★ For a sphere, surface area and volume are given as

$$A = 4\pi r^2, V = \frac{4}{3}\pi r^3$$

So, relative errors in them are given as

$$\frac{\Delta A}{A} = 2 \frac{\Delta r}{r} \text{ and } \frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$$

★ When two resistors R_1 and R_2 are connected

(i) In series $R_s = R_1 + R_2 \Rightarrow \Delta R_s = \Delta R_1 + \Delta R_2$

$$\Rightarrow \frac{\Delta R_s}{R_s} = \frac{\Delta R_1 + \Delta R_2}{R_1 + R_2}$$

(ii) In parallel,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{\Delta R_p}{R_p^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

★ $K.E = K = \frac{1}{2}mv^2$

$$\frac{\Delta K}{K} = \frac{\Delta m}{m} + 2 \frac{\Delta v}{v}$$

Q. If speed of a moving particle is increased by 3% then K.E of particle will increase by _____ %

Sol. $K = \frac{1}{2}mv^2$

$$\frac{\Delta k}{k} = \frac{\Delta m}{m} + 2 \frac{\Delta v}{v}$$

$$\frac{\Delta k}{k} = 2 \times 3\% = \underline{\underline{6\%}}$$

Q. Radius of sphere is measured to be $(7.50 \pm .85)$ cm. find % error in its volume

(JEE mains 2021 (18 march 2021))

Sol. $V = \frac{4}{3}\pi R^3$

$$\frac{\Delta v}{v} = 3 \frac{\Delta R}{R}$$

$$\frac{\Delta v}{v} = 3 \times \frac{.85}{7.50} = 3 \times \frac{.85}{7.5} \times 100$$

Q. If relative error for surface area of sphere of radius R is α . Find relative error for volume of sphere. (JEE mains 2018)

Sol. $A = 4\pi R^2$

$$\left(\frac{\Delta A}{A} \right) = 2 \frac{\Delta R}{R} = \alpha$$

$$\frac{\Delta R}{R} = \frac{\alpha}{2}$$

$$V = \frac{4}{3}\pi R^3$$

$$\frac{\Delta V}{V} = 3 \frac{\Delta R}{R}$$

$$\frac{\Delta v}{v} = 3 \cdot \frac{\alpha}{2} = \frac{3\alpha}{2}$$

- Q. The resistance $R = \frac{V}{I}$, where $V = (50 \pm 2)V$ and $I = (20 \pm 0.2)A$. The percentage error in R is 'x%'. The value of 'x' to the nearest integer is _____.

Jee Main 2021 (Online) 16th March. Ev. Shift

Sol. $R = \frac{V}{I} = VI^{-1}$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R}{R} = \frac{2}{50} + \frac{.2}{20} = \frac{.4 + .1}{10} = .05$$

$$\frac{\Delta R}{R} \times 100 = .05 \times 100 = 5$$

- Q. In the density measurement of a side ℓ , the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The fractional error in the measurement of density is:

Jee Main 2019 (Online) 9th April, Morning Shift

Sol. $\rho = \frac{m}{\text{vol}^n} = \frac{m}{l^3} = ml^{-3}$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 3 \frac{\Delta l}{l}$$

$$\frac{\Delta \rho}{\rho} = \frac{.1}{10} + 3 \times \frac{.01}{.1} = .01 + .3 = .31$$

$$\text{vol}^n \equiv \pi r^2 l$$

- Q. A wire has a mass 0.3 ± 0.003 g, radius 0.5 ± 0.005 mm and length 6 ± 0.06 cm. The maximum percentage error in the measurement of its density is [JEE 2004]

Sol. $\rho = \frac{m}{\pi r^2 l}$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 2 \frac{\Delta r}{r} + \frac{\Delta l}{l}$$

$$\frac{\Delta \rho}{\rho} = \frac{.003}{.3} + \frac{2 \times .005}{.5} + \frac{.06}{6}$$

$$\frac{\Delta \rho}{\rho} \times 100 = 1 + 2 \times 1 + 1 = 4\%$$

Ans. 4%

- Q. A particle is moving with speed $(20 \pm .1)$ m/s, find distance travelled by particle in $(5 \pm .2)$ sec.

Sol. Distance = $20 \times 5 = 100$

$$\text{Distance} = (100 \pm ?)$$

$$\text{Distance} = (S \pm \Delta S)$$

$$S = ut$$

$$\frac{\Delta S}{S} = \frac{\Delta v}{v} + \frac{\Delta t}{t}$$

$$\frac{\Delta S}{100} = \frac{.1}{20} + \frac{.2}{5}$$

$$\Delta S = .5 + .4 = .9$$

$$\text{distance} = (100 \pm .9)m$$

- Q. A particle moves with a constant velocity and covers a distance of (1000 ± 30) m in time (100 ± 2) s. Calculate the velocity of the particle with error limits.

Sol. velocity = $\frac{1000}{100} = 10 \text{ m/s}$ $V = (10.0 \pm .5)$

$$v = \frac{s}{t}$$

$$\frac{\Delta v}{v} = \frac{\Delta s}{s} + \frac{\Delta t}{t}$$

$$\frac{\Delta v}{10} = \frac{30}{1000} + \frac{2}{100}$$

$$\Delta V = .3 + .2 = .5$$

- Q. $Z = \frac{A^2}{B}$ (given) If $A \pm \Delta A = 100 \pm 0.5$ and $B \pm \Delta B = 10 \pm 0.1$. Calculate the value of Z with error limits.

Sol. Let $Z = (1000 \pm 20)$

$$Z = \frac{A^2}{B} = \frac{(100)^2}{10} = 1000$$

$$\frac{\Delta Z}{Z} = 2 \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$$\frac{\Delta Z}{1000} = 2 \frac{.5}{100} + \frac{.1}{10}$$

$$\Delta Z = 10 + 10 = 20$$

- Q. Two resistors of resistances $R_1 = 100 \pm 3$ ohm and $R_2 = 200 \pm 4$ ohm are connected (a) in series, (b) in parallel. Find the equivalent resistance of the (a) series combination. (b) Parallel combination. Use for (a) the relation $R = R_1 + R_2$

$$\text{and for (b)} \quad \frac{1}{R'} = \frac{1}{R_1} + \frac{1}{R_2}$$

Sol. (1) Series

$$R_{eq} = (300 \pm 7)$$

Parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{100} + \frac{1}{200}$$

$$R_{eq} = \frac{200}{3}$$

$$R_{eq} = \left(\frac{200}{3} \pm \frac{16}{9} \right)$$

(a) The equivalent resistance of series

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$-\frac{1}{R_{eq}^2} dR_{eq} = -\frac{1}{R_1^2} dR_1 - \frac{1}{R_2^2} dR_2$$

$$\frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\frac{\Delta R_{eq}}{\left(\frac{200}{3}\right)^2} = \left(\frac{3}{(100)^2} + \frac{4}{(200)^2} \right)$$

Q. Calculate focal length of a spherical mirror from the following observations. Object distance $u = (30.0 \pm 0.4)$ cm and image distance $v = (60.0 \pm 0.3)$ cm.

Sol. Spherical mirror formula is given by

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Differentiating w.r.t time we get

$$-\frac{1}{v^2} dv - \frac{1}{u^2} du = -\frac{1}{f^2} df$$

$$\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2} = \frac{\Delta f}{f^2}$$

$$\frac{.3}{60^2} + \frac{.4}{(30)^2} = \frac{\Delta f}{20^2}$$

$$\Delta f = 0.2$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{60} + \frac{1}{30} = \frac{1}{f}$$

$$\frac{1+2}{60} = \frac{1}{f}$$

$$f = 20$$

$$\frac{1}{f} = \frac{1}{+20} - \frac{1}{-10} = \frac{1}{20} + \frac{1}{10} = \frac{3}{20}$$

$$f = \frac{20}{3}$$

$$\text{Ans. } f = (20.0 \pm 0.2) \text{ cm}$$

Q. In an experiment for determination of the focal length of a thin convex lens, the distance of the object from the lens is 10 ± 0.1 cm and the distance of its real image from the lens is 20 ± 0.2 cm. The error in the determination of focal length of the lens is $n\%$. The value of n is _____ (JEE Advance 2023)

Sol. Lens formula is given by

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Differentiating wrt time we get

$$+\frac{1}{f^2} df = +\frac{1}{v^2} dv + \frac{1}{u^2} du$$

$$\frac{\Delta f}{f^2} = \frac{\Delta v}{v^2} + \frac{\Delta u}{u^2}$$

$$\Rightarrow \frac{\Delta f}{f} = f \left(\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2} \right)$$

$$= \frac{20}{3} \left(\frac{.2}{20 \times 20} + \frac{.1}{10 \times 10} \right)$$

$$= \frac{20}{3} \left(\frac{1}{2000} + \frac{1}{1000} \right)$$

$$= \frac{20}{3} \frac{3}{2000} = \frac{1}{100} \Rightarrow 1\%$$

Q. In an experiment of simple pendulum, the errors in the measurement of length of the pendulum (L) and time period (T) are 3% and 2% respectively. The maximum percentage error in the value of L/T^2 is

Sol. Maximum percentage in the value of L/T^2

$$= \frac{\Delta L}{L} \times 100\% + 2 \frac{\Delta T}{T} \times 100\%$$

$$= 3 + 2 \times 2 = 7\%$$

Q. A body travels uniformly a distance (13.8 ± 0.2) m in a time (4.0 ± 0.3) s. Calculate its velocity with error limits. What is the percentage error in velocity?

Sol. Given distance, $s = (13.8 \pm 0.2)$ m

and time $t = (4.0 \pm 0.3)$ s

$$\text{Velocity } v = \frac{s}{t} = \frac{13.8}{4.0} = 3.45 \text{ ms}^{-1} = 3.5 \text{ ms}^{-1}$$

$$\frac{\Delta v}{v} = \pm \left(\frac{\Delta s}{s} + \frac{\Delta t}{t} \right) = \pm \left(\frac{0.2}{13.8} + \frac{0.3}{4.0} \right) = \pm \left(\frac{0.8 + 4.14}{13.8 \times 4.0} \right)$$

$$\therefore \Delta v = \pm 0.0895 \times v = \pm 0.0895 \times 3.45 \\ = \pm 0.3087 = \pm 0.31$$

$$\text{Hence } v = (3.5 \pm 0.31) \text{ ms}^{-1}$$

Percentage error in velocity

$$= \frac{\Delta v}{v} \times 100 = \pm 0.0895 \times 100 = \pm 8.95\% = \pm 9\%$$

Q. The heat generated in a circuit is given by $Q = I^2 R t$, where I is current, R is resistance and t is time. If the percentage errors in measuring I , R and t are 2%, 1% and 1% respectively, then the maximum error in measuring heat will be

Sol. The percentage error in heat is given as

$$\frac{\Delta Q}{Q} \% = 2 \frac{\Delta I}{I} \% + \frac{\Delta R}{R} \% + \frac{\Delta t}{t} \%$$

Substituting the values, maximum possible percentage error is

$$= 2 \times 2\% + 1 \times 1\% + 1 \times 1\% = 6\%$$

Q. The following observations were taken for determining surface tension of water by capillary tube method: Diameter of capillary, $D = 1.25 \times 10^{-2}$ m and rise of water in capillary, $h = 1.45 \times 10^{-2}$ m.

Taking $g = 9.80 \text{ ms}^{-2}$ and using the relation

$$T = (rgh/2) \times 10^3 \text{ Nm}^{-1}, \text{ what is the possible error in surface tension } T?$$

Sol. Given $T = (rgh/2) \times 10^3 \text{ Nm}^{-1}$,

$$D = 1.25 \times 10^{-2} \text{ m}, h = 1.45 \times 10^{-2} \text{ m},$$

$$g = 9.80 \text{ ms}^{-2}$$

$$\frac{\delta T}{T} = \frac{\delta r}{r} + \frac{\delta h}{h} + \frac{\delta g}{g} \Rightarrow \frac{\delta T}{T} = \frac{0.01}{1.25} + \frac{0.01}{1.45} + \frac{0.01}{9.8}$$

after applying the above values in this relation we get $\delta T \% = 1.6\%$

4. Using screw gauge, the observation of the diameter of a wire are 1.324, 1.326, 1.334, 1.336 cm respectively. The value of diameter of wire along with its percentage uncertainty will be

Sol. Average diameter:

$$\bar{D} = \frac{\Sigma(D)}{N} = \frac{1.324 + 1.326 + 1.334 + 1.336}{4} = 1.330$$

$$\Delta D_1 = 1.324 - 1.330 = -0.006$$

$$\Delta D_2 = 1.326 - 1.330 = -0.004$$

$$\Delta D_3 = 1.334 - 1.330 = 0.004$$

$$\Delta D_4 = 1.336 - 1.330 = 0.006$$

Mean absolute error:

$$\overline{\Delta D} = \frac{|\Delta D_1| + |\Delta D_2| + |\Delta D_3| + |\Delta D_4|}{4}$$

$$= \frac{0.006 + 0.004 + 0.004 + 0.006}{4} = \frac{0.020}{4} = 0.005 \text{ cm}$$

$$\text{Relative error} = \frac{\overline{\Delta D}}{D} = \frac{0.005}{1.330} = 0.004$$

$$\% \text{ error} = \frac{\overline{\Delta D}}{D} \times 100 = 0.4\%$$

Q. The length of a rectangular plate is measured by a meter scale and is found to be 10.0 cm. Its width is measured by vernier callipers as 1.00 cm. The least count of the meter scale and vernier callipers are 0.1 cm and 0.01 cm respectively (Obvious from readings). Maximum permissible error in area measurement is

Sol. $A = \ell b = 10.0 \times 1.00 = 10.00$

$$\text{Now, } \frac{\Delta A}{A} = \frac{\Delta \ell}{\ell} + \frac{\Delta b}{b}$$

$$\frac{\Delta A}{10.00} = \frac{0.1}{10.0} + \frac{0.01}{1.00}$$

$$\Rightarrow \Delta A = 10.00 \left(\frac{1}{100} + \frac{1}{100} \right) = 10.00 \left(\frac{2}{100} \right) = \pm 0.2 \text{ cm}^2$$

Q. To estimate 'g' (from $g = 4\pi^2 \frac{L}{T^2}$), error in measurement of L is + 2% and error in measurement of T is + 3%. The error in estimated 'g' will be

$$\text{Sol. } g = 4\pi^2 \frac{L}{T^2}$$

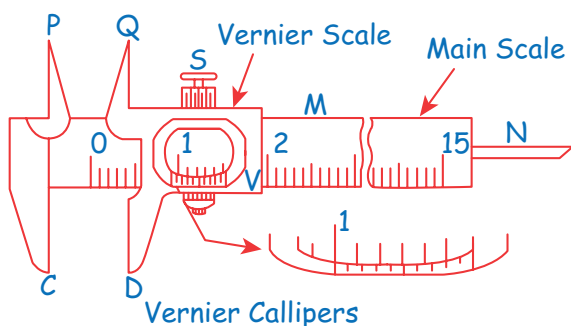
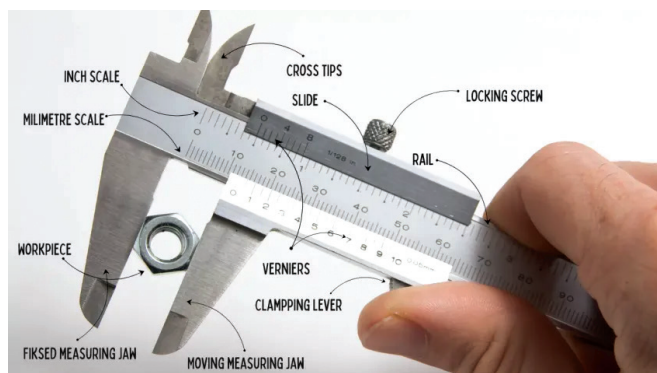
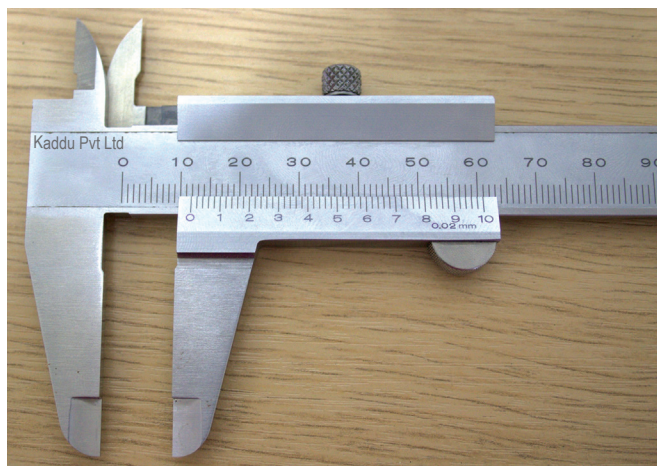
$$\frac{\Delta \ell}{\ell} = 2\% = \pm 2 \times 10^{-2}$$

$$\frac{\Delta T}{T} = +3\% = \pm 3 \times 10^{-2}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T} = 2 \times 10^{-2} + 2 \times 3 \times 10^{-2}$$

$$= 8 \times 10^{-2} = \pm 8\%$$

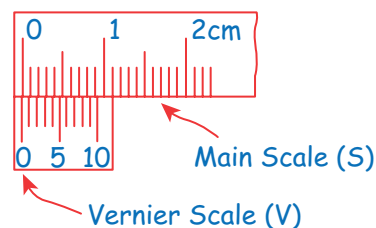
VERNIER CALLIPERS



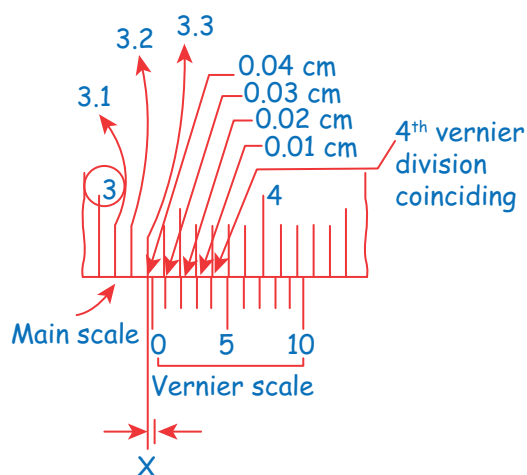
PRINCIPLE OF VERNIER

The principle of a vernier caliper relies on the use of two graduated scales, a main scale and a vernier scale, to achieve more accurate measurements than a standard ruler.

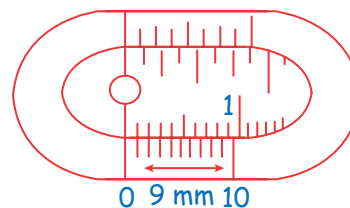
The caliper utilizes a main scale, which is similar to a ruler, and a vernier scale that slides along it.



Here we need to find the value of x which can be calculated x as shown in the figure. Here 4th division on vernier scale coincides with main scale division value of x can be calculated from the following diagram as shown in the figure.



Principle of Vernier



★ N divisions on the vernier scale is equal to $(N-1)$ divisions on the main scale.

★ $N \text{ VSD} = (N - 1) \text{ M.S.D.}$

$$1 \text{ VSD} \rightarrow \left(\frac{N-1}{N} \right) \text{ MSD}$$

$$\text{Least Count} = \text{LC} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 1 \text{ MSD} - \left(\frac{N-1}{N} \right) \text{ MSD}$$

$$L.C = \frac{1 \text{ MSD}}{N}$$

Suppose we have vernier calliper whose 9 main scale division MSD is divided into 10 vernier scale division VSD (Mostly yahi vernier calliper hum lab me use karte hain aur exam me bhi jyadatar yahi given hota hai but advance me ye data vary hota rahta hai to pahle hum conceptually ise samjhte

hain baki baad me padhenge.) $L.C = \frac{1MSD}{N} \Rightarrow (1$

$MSD = 1 \text{ mm and } N = 10)$

So $LC = 1 \text{ mm}/10 = 0.1 \text{ mm} = 0.01 \text{ cm}$

or $10 \text{ VSD} = 9 \text{ MSD}$

So $1 \text{ VSD} = 9/10 \text{ MSD}$

$L.C. = 1 \text{ MSD} - 1 \text{ VSD}$

$$= 1 \text{ MSD} - \frac{9}{10} \text{ MSD} = 0.1 \text{ MSD}$$

$$L.C = \frac{1}{10} \text{ MSD}$$

$$1 \text{ MSD} = 1 \text{ mm}$$

$$L.C = \frac{1}{10} (1 \text{ mm})$$

$$L.C = \frac{1}{10} \text{ mm} = 0.1 \text{ mm} = 0.01 \text{ cm}$$

Q. A Vernier calipers has 1 mm marks on the main scale. It has 20 equal divisions on the Vernier scale which match with 16 main scale divisions. For this Vernier, the least count is [JEE 2010]

Sol. $20 \text{ V.S.D} \rightarrow 16 \text{ M.S.D}$

$$1 \text{ VSD} \rightarrow \frac{16}{20} \text{ MSD}$$

$$L.C = 1 \text{ M.S.D} - 1 \text{ V.S.D}$$

$$= 1 \text{ M.S.D} - \frac{16}{20} \text{ M.S.D}$$

$$= \frac{1}{5} \text{ M.S.D} = \frac{1}{5} \text{ mm}$$

Q. The Vernier of a circular scale is divided into 30 divisions, which coincides with 29 main scale divisions. If each main scale division is $(\frac{1}{2})^\circ$, the least count of the instrument is

Sol. $30 \text{ V.S.D} \rightarrow 29 \text{ M.S.D}$

$$1 \text{ VSD} \rightarrow \frac{29}{30} \text{ MSD}$$

$$L.C = 1 \text{ M.S.D} - 1 \text{ V.S.D} = 1 \text{ MSD} - \frac{29}{30} \text{ MSD}$$

$$L.C = \frac{1}{30} \left(\frac{1}{2} \right)^\circ = \left(\frac{1}{60} \right)^\circ = 1'$$

Q. A vernier callipers has its main scale of 10 cm equally divided into 200 equal parts. Its vernier scale of 25 divisions coincides with 12 mm on the main scale. The least count of the instrument is

Sol. 10 cm divided in 200 divisions, so

$$1 \text{ division} = \frac{10}{200} = 0.05 \text{ cm.}$$

$$\text{Now, } 25V = 24S.$$

$$\Rightarrow V = \frac{24}{25} S$$

$$LC = S - V$$

$$= S - \frac{24}{25} S = \frac{1}{25} S$$

$$\therefore 1S = 0.05 \text{ cm, thus}$$

$$\text{So, } LC = \frac{0.05}{25} = 0.002 \text{ cm}$$

Q. In an experiment the angles are required to be measured using an instrument 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half-a-degree ($= 0.5^\circ$), then the least count of the instrument is: [AIEEE - 2009]

Sol. $30 \text{ V.S.D} \rightarrow 29 \text{ M.S.D}$

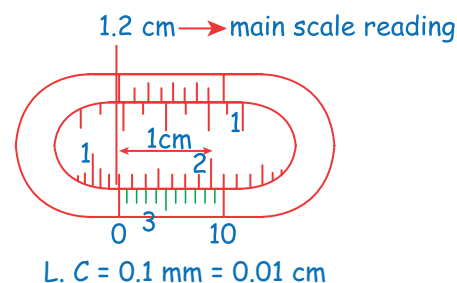
$$1 \text{ V.S.D} \rightarrow \frac{29}{30} \text{ MSD}$$

$$L.C = 1 \text{ MSD} - \frac{29}{30} \text{ MSD} = \frac{1}{30} \times 0.5^\circ = \frac{1}{60}$$

LENGTH AS MEASURED BY VERNIER CALLIPERS

★ The formula for measuring the length is $L = \text{main scale reading} + \text{least count of vernier scale} \times \text{Vernier scale division coinciding with a main scale division} - (\text{zero error})$

How to Take Reading of a Vernier scale



- ★ Main Scale reading = 1.2 cm ;
- ★ Vernier scale reading = $3 \times LC$
 $= 3 \times 0.01 \text{ cm} = 0.03 \text{ cm}$
- ★ So, Reading = $1.2 + 0.03 = 1.23 \text{ cm}$

SKC

Vernier calliper me reading lene ke liye hume carefully 3 steps follow karne hai

1. sabse pahle main scale reading note karo be carefull vernier scale ke zero wale dande ke left wali reading leni hai.

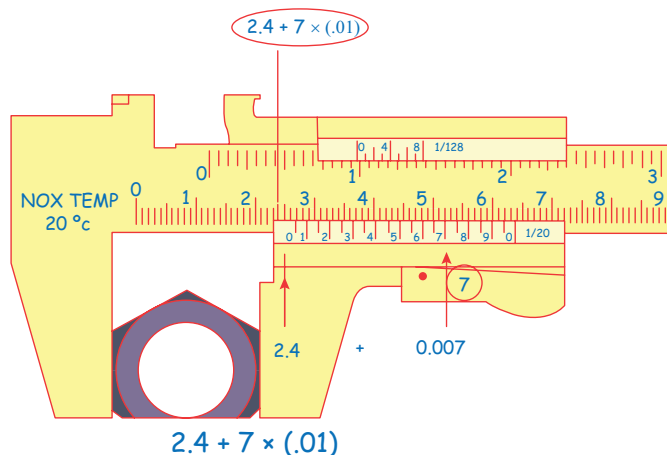
2. Ab ye dekho ki vernier scale ka kaun sa danda seedha direct main scale ke dande se coincide kar rha hai. Suppose 4th division coincide kar rha hai to vernier scale ki reading hui

3. Now reading will be = main scale reading + (concoiding division $\times$ LC) when no zero error

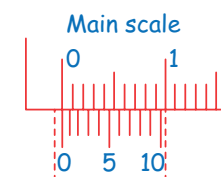
★ If zero error is there then now reading will be = main scale reading + (concoiding division $\times$ LC) - (zero error with sign)

SKC ki ye line yaad rakhna zero error ko sign ke sath ghatana hai (Substract zero error with sign)

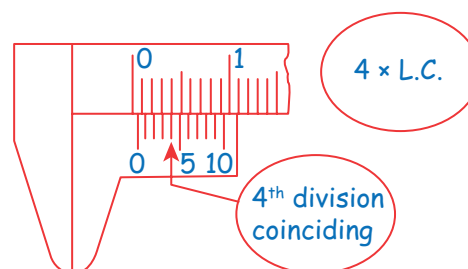
Another Example



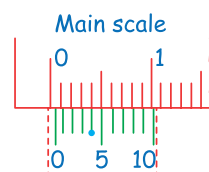
Zero Errors



Vernier scale without zero error

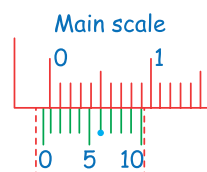


Positive zero error (+0.04 cm) and its correction

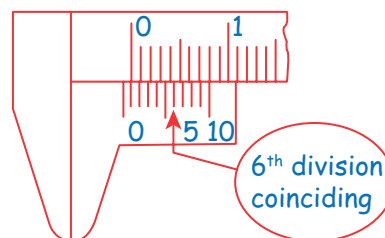


Vernier scale with positive zero error (I)

Here zero of vernier scale is right side (ahead) of main scale and 4th division of vernier scale is coinciding with main scale hence zero error = $+4 \times LC$



Vernier scale with negative zero error (I)



Negative zero error = (-0.04 cm) and its correction

Negative zero error = $-(10 - 6) \times LC$
 $= -4 \times LC = -0.04 \text{ cm}$

Reading = (main scale reading) + (coinciding division $\times$ L.C.) - (Z.E with sign)

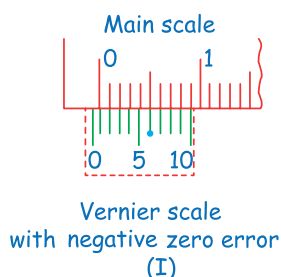
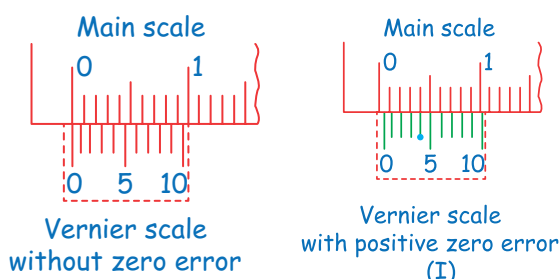
Bhai, hasna mat, yaad karne ki badiya trick hai ki agar negative zero error ho, to vernier scale ki reading ke liye peeche se danda count karo.



ZERO ERROR

Types of Zero Error in vernier calipers

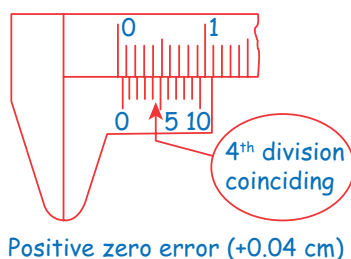
- Negative Zero error
- Positive Zero error



Positive Zero Error

When the jaws of the vernier calipers are closed, and if the zero mark on the auxiliary (vernier) scale lies to the right of the main scale zero-mark, then error is called Positive Zero error.

$$Z.E = +4 \times (.01) \text{ cm}$$



Q. Least count of a vernier calipers is 0.01 cm. When the two jaws of the instrument touch each other, the 5th division of the vernier scale coincide with a main scale division and the zero of the vernier scale lies to the left (of the zero of the main scale). Furthermore, while measuring the diameter of a sphere, the zero mark of the vernier scale lies between 3.2 cm and 3.3 cm and the 7th vernier division coincides with a main scale division. Calculate the diameter of the sphere.

- ve zero error

$$\text{Sol. } Z.E = -5 \times .01 = -.05 \text{ cm}$$

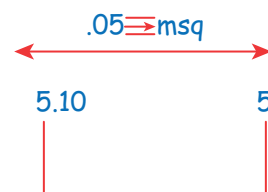
$$\begin{aligned} \text{Reading} &= 3.2 + (7 \times .01) - (-.05) \text{ cm} \\ &= 3.2 + .12 = \underline{3.32 \text{ cm}} \end{aligned}$$

Q. The diameter of a cylinder is measured using a Vernier callipers with no zero error. It is found that the zero of the Vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The Vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is: [JEE-Advance 2013]

$$\text{Sol. Main scale} \rightarrow 5.10 \quad 50 \text{ V.S.D} \rightarrow 2.45 \text{ cm}$$

$$\begin{aligned} 5.10 \text{ cm} + 24 \times (\text{L.C}) - 0.1 \text{ V.S.D} &\rightarrow \frac{2.45}{50} \\ &= .049 \text{ cm} \end{aligned}$$

$$\begin{aligned} 5.10 + 24 \times .001 & \quad \text{L.C.} = 1 \text{ MSD} - 1 \text{ VSD} \\ 5.10 + .024 & = .05 \text{ cm} - .049 \text{ cm} \\ = \underline{5.124 \text{ cm}} & = .001 \text{ cm} \end{aligned}$$



Q. A spectrometer gives the following reading when used to measure the angle of a prism. Main scale reading : 58.5 degree. Vernier scale reading: 09 divisions. Given that 1 division on main scale corresponds to 0.5 degree. Total divisions on the vernier scale is 30 and match with 29 divisions of the main scale. The angle of the prism from the above data:

$$LC = \left(\frac{1}{60} \right)^\circ \quad [\text{AIEEE} - 2012]$$

$$58.5 + 9 \times \frac{1}{60} = 58.5 \quad 30 \text{ VSD} \rightarrow 29$$

$$\quad \quad \quad .15 \quad \rightarrow \frac{29}{30}$$

$$\quad \quad \quad \underline{58.65}$$

Q. The main scale of a vernier calipers reads in millimeter and its vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. When the two jaws of the instrument touch each other the seventh division of the vernier scale coincide with a scale division and the zero of the vernier lies to the right of the zero of main scale. Furthermore, when a cylinder is tightly placed along its length between the two jaws, the zero of the vernier scale lies slightly to the left of 3.2 cm; and the fourth vernier division coincides with a scale division. Calculate the measured length of the cylinder.

Sol. Z.E = +7 (L.C) main scale $\rightarrow$ 3.1 cm

$$3.1 + 4 \times (.01 \text{ cm}) - (7 \times (.01) \text{ cm})$$

$$= 3.1 + .04 - .07 = 3.07 \text{ cm}$$

Q. The diameter of a sphere is measured using a vernier caliper whose 9 divisions of main scale are equal to 10 divisions of vernier scale. The shortest division on the main scale is equal to 1 mm. The main scale reading is 2 cm and second division of vernier scale coincides with a division on main scale. If mass of the sphere is 8.635 g, the density of the sphere is:

[08 April, 2024 (Shift-I)]

Sol. Here, 9MSD = 10VSD and mass = 8.635 g

For least count we have,

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$LC = 1 \text{ MSD} - \frac{9}{10} \text{ MSD} = \frac{1}{10} \text{ MSD}$$

$$LC = 0.01 \text{ cm}$$

$$\text{Reading of diameter} = \text{MSR} + LC \times \text{VSR}$$

$$= 2 \text{ cm} + (0.01) \times (2) = 2.02 \text{ cm}$$

Volume of sphere

$$= \frac{4}{3} \pi \left(\frac{d}{2} \right)^3 = \frac{4}{3} \pi \left(\frac{2.02}{2} \right)^3 = 4.32 \text{ cm}^3$$

Density

$$= \frac{\text{mass}}{\text{volume}} = \frac{8.635}{4.32} = 1.998 \sim 2.00 \text{ g}$$

Q. In a vernier calliper, when both jaws touch each other, zero of the vernier scale shifts towards left and its 4th division coincides exactly with a certain division on main scale. If 50 vernier scale divisions equal to 49 main scale divisions and zero error in the instrument is 0.04 mm then how many main scale divisions are there in 1 cm? [06 April, 2024 (Shift-II)]

Sol. 50 VSD = 49 MSD

$$1 \text{ VSD} = \left(\frac{49}{50} \right) \text{ MSD}$$

$$L.C. = .01 \text{ mm} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$\Rightarrow .01 \text{ mm} = 1 \text{ MSD} - \left(\frac{49}{50} \right) \text{ MSD}$$

$$\Rightarrow 1 \text{ MSD} = 0.5 \text{ mm}$$

$$\therefore \text{No. of main scale divisions in 1 cm} = \frac{10 \text{ mm}}{.5 \text{ mm}} = 20$$

Q. A vernier callipers has 20 divisions on the vernier scale, which coincides with 19th division on the main scale. The least count of the instrument is 0.1 mm. One main scale division is equal to ____ mm. [05 April, 2024 (Shift-II)]

Sol. 20 VSD = 19 MSD

$$L.C. = 0.1 \text{ mm} = \text{MSD} - 1 \text{ MSD}$$

$$.01 \text{ mm} = 1 \text{ MSD} - \frac{19}{20} \text{ MSD} = \frac{1 \text{ MSD}}{20}$$

$$1 \text{ MSD} = 2 \text{ mm}$$

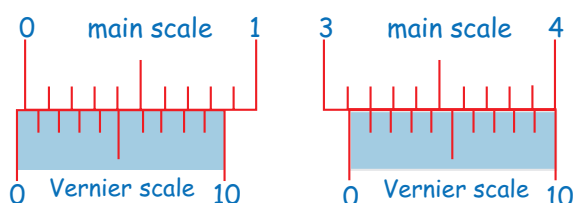
Q. The vernier scale used for measurement has a positive zero error of 0.2 mm. If while taking a measurement it was noted that zero on the vernier scale lies between 8.5 cm and 8.6 cm. and vernier coincidence is 6, then the correct value of measurement is _____ (least count = 0.01 cm)

[17 March, 2021 (Shift-I)]

Sol. Reading = MSR + VSD $\times$ LC - zero error

$$\text{Reading} = 8.5 + \frac{(0.1) \times 6}{10} - \frac{0.2}{10} = 8.54$$

- Q.** The smallest division on the main scale of a Vernier callipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this calliper with no gap between its two jaws. The figure on the right shows the reading with a solid sphere held between the jaws. The correct diameter of the sphere is [JEE Adv, 2021]



Sol. Given 10 VSD = 9 MSD

$$\text{Least count} = 1 \text{ VSD} = \frac{9}{10} \text{ MSD}$$

$$\text{Least count} = 1 \text{ MSD} - 1 \text{ VSD} = \left(1 - \frac{9}{10}\right) \text{ MSD}$$

$$= 0.1 \text{ MSD} = 0.1 \times 0.1 \text{ cm} = 0.01 \text{ cm}$$

As '0' of V.S. lie before '0' of M.S.

$$\text{Zero error} = -[10 - 6] \text{ L.C.}$$

$$= -4 \times 0.01 \text{ cm}$$

$$= -0.04 \text{ cm}$$

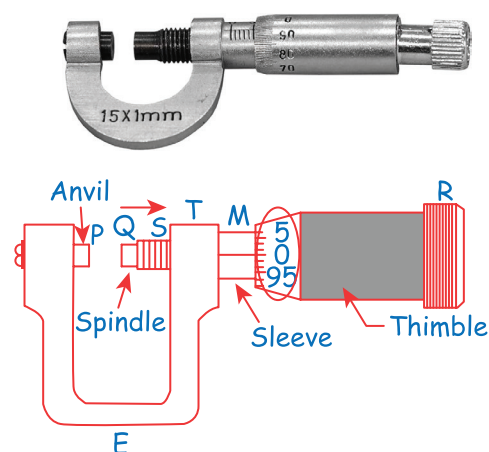
$$\text{Reading} = 3.1 \text{ cm} + 1 \times \text{LC} = 3.4 \text{ cm} + 1 \times 0.01 \text{ cm}$$

$$= 3.11 \text{ cm}$$

$$\text{True diameter} = \text{Reading} - \text{Zero error}$$

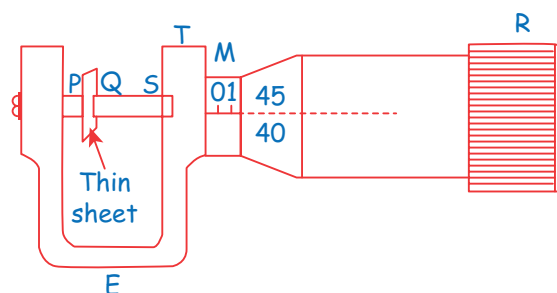
$$= 3.11 - (-0.04) \text{ cm} = 3.15 \text{ cm}$$

SCREW GAUGE

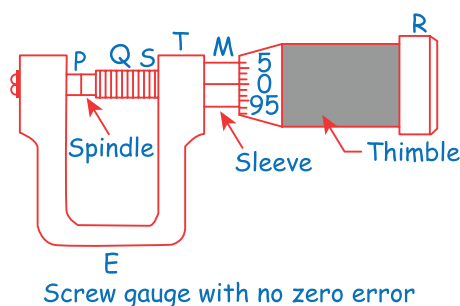


Principle of a Micrometer

The screw gauge operates on the principle of a screw moving linearly within a nut, converting rotational motion into precise axial movement. This linear movement is then measured using a graduated scale and a circular scale (thimble), allowing for accurate measurement of small dimensions like the diameter of a thin wire or the thickness of a thin sheet.

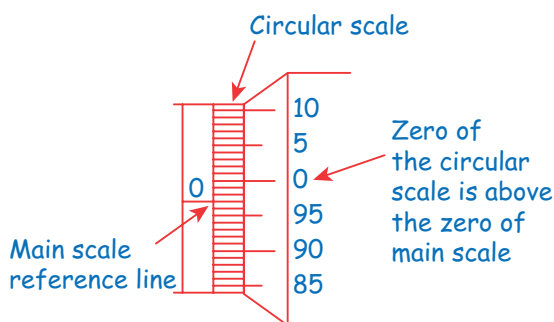


No Zero Error



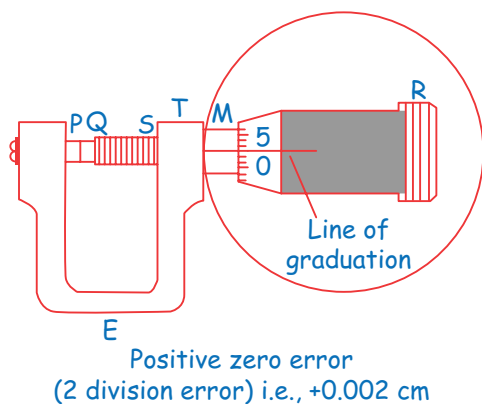
Negative Zero Error

The zero mark on the circular scale lies above the zero mark on the main scale. In this case, the screw gauge will give a reading that is lower than the actual value when the anvil and spindle are closed.



Positive Zero Error

The zero mark on the circular scale lies below the zero mark on the main scale. In this case, the screw gauge will give a reading that is higher than the actual value when the anvil and spindle are closed.



12 mm → main scale reading

$$L.C = \frac{\text{pitch}}{\text{No. of div. on circular scale}}$$

100 dande → 1 mm → pitch

100 dande → $\frac{1}{100}$ mm → L.C

20 dande → $\frac{20}{100}$ mm

Reading = M.S.R + (coincidin div) × L.C. - (Z.E.)

$$L.C = \frac{\text{pitch}}{\text{No. of div. on circular scale}}$$

CONSTANTS OF THE SCREW GAUGE

(a) **Pitch:** The translational motion of the screw is directly proportional to the total rotation of the head. The pitch of the instrument is the distance between two consecutive threads of the screw which is equal to the distance moved by the screw due to one complete rotation of the cap. Thus for 10 rotation of cap distance moved by screw 5 mm, then pitch = 0.5 mm

(b) **Least count:** In this case also, the minimum (or least) measurement (or count) of length is equal to one division on the head scale which is equal to pitch divided by the total cap divisions. Thus is the aforesaid Illustration, if the total cap division is 100, then least count = 0.5 mm/100 = 0.005 mm

(c) Measurement of length by screw gauge:

$$L = n \times \text{pitch} + f \times \text{least count} - (\text{zero error}),$$

where n = main scale reading and f = caps scale reading

Q. The pitch of a screw gauge is 0.5 mm and there are 50 divisions on the circular scale. In measuring the thickness of a metal plate, there are five divisions on the pitch scale (or main scale) and thirty fourth division coincides with the reference line. Calculate the thickness of the metal plate.

Sol. Pitch = .5 mm 50 → .5 mm

$$L.C. = \frac{.5 \text{ mm}}{50} = \frac{1}{100} \text{ mm} = .01 \text{ mm} \quad 1 \rightarrow .5$$

$$\begin{aligned} \text{Reading} &= 5 \times .5 \text{ mm} + 34 \times L.C. \quad 284 \rightarrow \frac{.5}{50} \times 284 \\ &= 2.5 \text{ mm} + 34 \times .01 \\ &= 2.84 \text{ mm} \end{aligned}$$

Q. The pitch of a screw gauge is 1 mm and there are 100 divisions on its circular scale. When a wire is placed between the jaws, 3 linear scale divisions are visible while 56 divisions on circular scale coincide with the reference line. Determine the diameter of the wire.

$$\text{Sol. Reading} = 3 \times \underbrace{1 \text{ mm}}_{\text{pitch}} + \underbrace{56}_{\text{coincide}} \times \underbrace{\left(\frac{1 \text{ mm}}{100}\right)}_{L.C.}$$

Ans: 3.56 mm

Q. A screw gauge gives the following reading when used to measure the diameter of a wire. Main scale reading : 0 mm. Circular scale reading: 52 divisions Given that 1 mm on main scale corresponds to 100 divisions of the circular scale. The diameter of wire from the above date is: [AIEEE - 2011]

$$\text{Sol. } 0 + 52 \times \frac{1}{100} \text{ mm} \left(L.C = \frac{1}{100} \text{ mm} \right)$$

- Q.** The pitch of a screw gauge is 0.5 mm and there are 100 divisions on its circular scale. The instrument reads +2 divisions when nothing is put in-between its jaws. In measuring the diameter of a wire, there are 8 divisions on the main scale and 83rd division coincides with the reference line. Then the diameter of the wire is

Sol. $\Delta l = 0.5 \text{ mm}$

$N = 100 \text{ divisions}$

zero correction = 2 divisions

Reading = Measured value - zero correction

$$= (8 \times 0.5 \text{ mm}) + (83) \times \frac{0.5}{100} \text{ mm} - 2 \times \frac{0.5}{100}$$

$$= 4 \text{ mm} + 0.415 \text{ mm} - 0.01 \text{ mm}$$

$$= 4.405 \text{ mm}$$

- Q.** The pitch of a screw gauge having 50 divisions on its circular scale is 1 mm. When the two jaws of the screw gauge are in contact with each other, the zero of the circular scale lies 6 divisions below the line of graduation. When a wire is placed between the jaws, 3 linear scale divisions are clearly visible while 31st division on the circular scale coincides with the reference line. The diameter of the wire is

Sol. $\Delta l = 1 \text{ mm}$

$N = 50 \text{ division}$

zero error = -6 divisions = -0.12 mm

Diameter = Measured value - zero correction

$$= 3 \times 1 \text{ mm} + (31) \times \frac{1}{50} \text{ mm} - (-0.12 \text{ mm})$$

$$= 3 + 0.62 \text{ mm} + 0.12 \text{ mm} = 3.74 \text{ mm}$$

- Q.** The smallest division on the main scale of a vernier callipers is 1 mm, and 10 vernier divisions coincide with 9 main scale divisions. While measuring the diameter of a sphere, the zero mark of the vernier scale lies between 2.0 and 2.1 cm and the fifth division of the vernier scale coincides with a mark on main scale. Then diameter of the sphere is

Sol. $D = 2 \times 1 \text{ cm} + 5 \times \left(\frac{10-9}{100} \right) \times \frac{1}{10} \text{ cm}$
 $= 2.05 \text{ cm}$

- Q.** The vernier constant of a vernier callipers is 0.1 mm and it has zero error of -0.05 cm. While measuring the diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The correct diameter will be $n \times 10^{-2} \text{ cm}$. Find n .

Sol. Measured Diameter = MSR + VSR $\times$ V.C

$$= 1.7 \text{ cm} + 0.01 \text{ cm} \times 5$$

$$= 1.75 \text{ cm}$$

Corrected Reading

$$= \text{Measured value} - \text{Zero error}$$

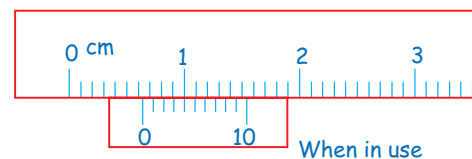
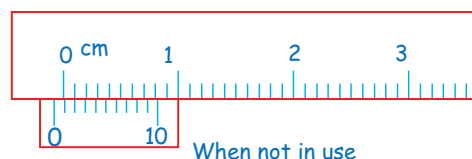
$$= 1.75 - (-0.05)$$

$$= 1.80 \text{ cm}$$

$$= 180 \times 10^{-2} \text{ cm}$$

$$\therefore n = 180$$

- Q.** The main scale of a Vernier callipers reads in millimeter and its vernier scale is divided into 10 divisions which coincide with 9 divisions of the main scale. The length of the object for the situation shown below is found to be $\frac{12x}{10} \text{ mm}$. Find the value of x .



Sol. Negative zero error = $(10 - 2) \times 0.1 \text{ mm}$
 $= 0.8 \text{ mm}$
 Reading = $6 \text{ mm} + 4 \times 0.1 \text{ mm} + 0.8 \text{ mm}$
 $= 7.2 \text{ mm}$
 $= \frac{12 \times 6}{10} \text{ mm} \Rightarrow x = 6$

Ab main kuch images add kar raha hoon, taaki aapki speed aur understanding badh sake...Ismein coinciding division mention kiya gaya hai, usi according answer dena.

